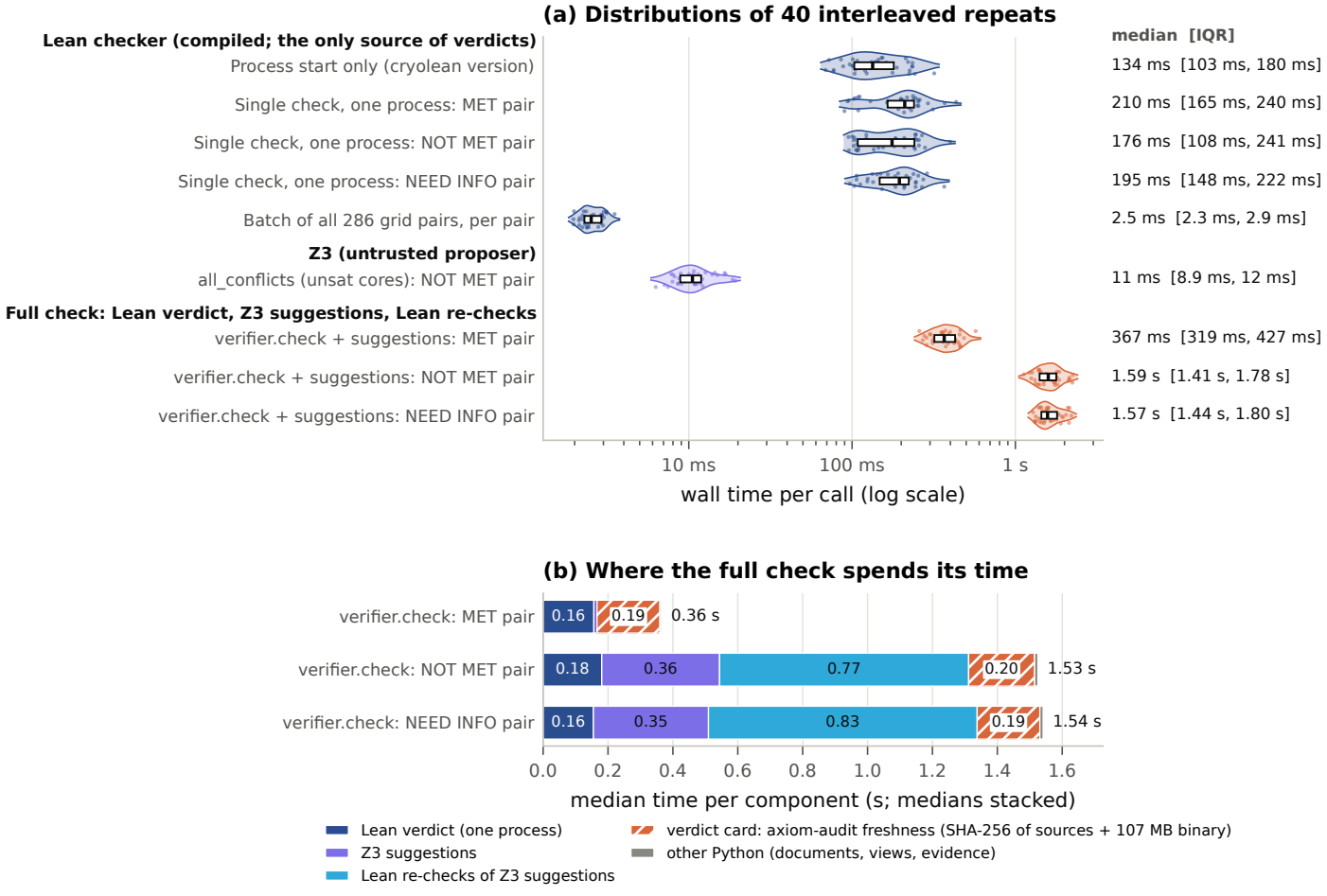

(a) Distributions of 40 interleaved repeats
Lean checker (compiled; the only source of verdicts)
Process start only (cryolean version)
Single check, one process: MET pair
Single check, one process: NOT MET pair
Single check, one process: NEED INFO pair
Batch of all 286 grid pairs, per pair
Z3 (untrusted proposer)
all_conflicts (unsat cores): NOT MET pair
Full check: Lean verdict, Z3 suggestions, Lean re-checks
verifier.check + suggestions: MET pair
verifier.check + suggestions: NOT MET pair
verifier.check + suggestions: NEED INFO pair
median [IQR]
134 ms [103 ms, 180 ms]
210 ms [165 ms, 240 ms]
176 ms [108 ms, 241 ms]
195 ms [148 ms, 222 ms]
2.5 ms [2.3 ms, 2.9 ms]
11 ms [8.9 ms, 12 ms]
367 ms [319 ms, 427 ms]
1.59 s [1.41 s, 1.78 s]
1.57 s [1.44 s, 1.80 s]
10 ms
100 ms
1 s
wall time per call (log scale)
(b) Where the full check spends its time
verifier.check: MET pair
0.16
0.19
0.36 s
verifier.check: NOT MET pair
0.18
0.36
0.77
0.20
1.53 s
verifier.check: NEED INFO pair
0.16
0.35
0.83
0.19
1.54 s
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
median time per component (s; medians stacked)
Lean verdict (one process)
Z3 suggestions
Lean re-checks of Z3 suggestions
verdict card: axiom-audit freshness (SHA-256 of sources + 107 MB binary)
other Python (documents, views, evidence)